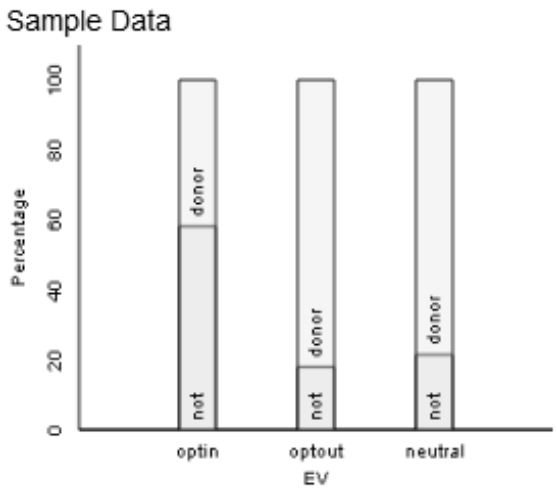
Solutions to Exploration 8.1

1. Observational units: volunteers
   Explanatory variable: recruiting strategy (opt in/opt out/neutral) categorical
   Response variable: organ donor (Y/N) categorical
2. Only random assignment was used. This means cause-and-effect conclusions can be made. Generalizing to a larger population should be done with caution since a random sample was not used to obtain the people in the study—they were volunteers.
3. No pairing was used in the study design; individuals from the different recruiting groups are independent of each other.
4. a. H_0: There is no association between recruiting strategy and organ donation.
   H_a: There is an association between recruiting strategy and organ donation.
   b. H_0: $\pi_{in} = \pi_{out} = \pi_{neutral,}$ π is the proportion of organ donors in the group.
   H_a: At least one π is different.
5. a.

	Opt in	Opt out	Neutral
Organ donor	23	41	44
Not organ donor	32	9	12
Total	55	50	56

   b. Opt out: 0.82 Opt in: 0.418 Neutral: 0.786

c.



d. Opt-out and neutral groups have higher proportions of donors compared to the opt-in group.

6. It is possible there is not a difference in the population proportions of donors recruited by each of these strategies and the proportions in our data happened by random chance alone. It is also possible that there is a difference in the population proportions of donors recruited by each strategy and that is why our data yielded different proportions.
7. a. Calculate all pairwise differences. Then find the average of the absolute value of the differences.

   b. Large values of this statistic would provide string evidence against the null hypothesis. Skewed right; MAD can never be negative.
8. 0.82 – 0.418 = 0.402; not a great choice because it ignores some of the groups.
9. a. Opt out: 0.82, opt in: 0.418, neutral: 0.786
   b. Opt out – opt in: 0.402, opt out – neutral: 0.034, opt in – neutral: -0.368; p-value is 0.
   c. MAD = 0.268
10. 108 black cards to represent the donors and 53 red cards to represent the nondonors. Shuffle the cards and deal 55 to the opt-in group, 50 to the opt-out group, and 56 to the neutral group. Calculate the proportion of black cards in each group to calculate the

simulated MAD statistic. Repeat several times to get an idea of MAD values to expect under the null of no association.

11. a. Yes, the simulated MAD is closer to zero than the observed MAD.
    b. Distribution is skewed to the right because the MAD statistic is always nonnegative. Approximate mean of 0.074 and SD of 0.038.
12. The p-value = 0
13. The p-value provides very strong evidence against the null and in support of an association between the recruiting strategy and organ donation.
14. Volunteers were from general interest bulletin boards on the Internet. They were offered a $4 incentive to participate. Since this was a self-selected sample and not a random sample it will be difficult to generalize the finding to a broader population.
15. We can draw cause-and-effect conclusions, since subjects were randomly assigned to one of the three recruiting strategies.
16. Researchers wanted to know if there was a recruitment strategy that was best for getting organ donors. 161 volunteers from general interest boards on the Internet were paid a $4 incentive to participate. The subjects were randomly assigned to one of three recruiting strategies: opt in, opt out, neutral. 41.8% of the opt-in group became organ donors, 82% of the opt-out group became donors, and 78.6% of the neutral group became donors. The MAD statistic of 0.268 was calculated from the data. A simulation-based analysis was carried out and found a p-value of zero. This offered strong evidence against the null and in support of there being an association between recruiting strategy and organ donation. A follow-up analysis calculated 95% confidence intervals for each pair of differences. From these confidence intervals we concluded that the opt-out and neutral groups had significantly higher proportions of organ donors than the opt-in group. The population proportion of organ donors were not found to be significantly different between the opt-out and neutral groups. We can only generalize these results to those who frequent online general interest bulletin boards and would volunteer for this kind of a study. We can conclude that the cause of the different organ donation rates was a cause of the group they were placed in. Further studies would actually place those renewing their licenses in one of the three groups.

Solutions to Exploration 8.2A

1. Do different socially normed signs produce different rates of towel reuse?
2. Explanatory: message type, categorical five categories

   Response: towel reuse, categorical two categories
3. Study design did not use pairing, the subjects who receive different signs are independent of each other.
4. H_0: There is no association between type of message and towel reuse.

   H_a: There is an association between message type and towel reuse.

   H_0: $\pi_{none} = \pi_{guest} = \pi_{room} = \pi_{citizen} = \pi_{gender}$.

   H_a: At least one π is different.
5. a.

Message type	None	Guest	Room	Citizen	Gender
Proportion reuse towel	0.37	0.441	0.493	0.434	0.410

   b. Largest proportion towel reuse is room, smallest is none.

   c. Yes, will probably find a difference in proportions of reuse for room and none.
6. a. MAD statistic = 0.055.

   b. The p-value = 0.034.

   c. We find strong evidence against the null and in favor of the alternative that at least one type of message results in a proportion of towel reuse that is different from the others.
7. (113 + 151 + 145 + 127 + 150)/1,595 = 686/1,595 = 0.43
8. None: 0.028, same: 0.028, citizen: 0.027, gender: 0.028, guest: 0.027
9. (0.37 – 0.43)/0.028 = –2.10 (none), (0.493 – 0.43)/0.028 = 2.24 (same), (0.434 – 0.43)/0.027 = 0.15 (citizen), (0.41 – 0.43)/0.028 = –0.73 (gender), (0.44 – 0.43)/0.027 = 0.41 (guest)
10. None and same room
11. Chi-square = $(-2.1)^2 + 2.24^2 + 0.15^2 + (-0.73)^2 + 0.41^2 = 10.15$
12. Small/positive if null is true; large/positive if alternative is true
13. The p-value = 0.036

14. The p-values are similar. The conclusions regarding strength of evidence against the null are the same.
15. Theory-based chi-square test p-value is 0.0379. The theory-based p-value is similar to the theory-based chi-square statistics, which leads to a similar conclusion.
16. a. There are 10 cells in the 2×2 table.

    b. All 10 cells have counts of at least 10, so validity conditions are met.
17. a. Ten

    b. None vs. same room and same room vs. gender show significant differences in the proportion of who reuses towels. The decision is based on whether or not zero is in the interval. If zero is in the interval for the difference in proportion of reusers, then it is plausible the population proportions are the same.

    c. The population proportion of guests who reuse their towel from the same room sign rooms is 0.045–0.201 higher than the population proportion of guests that reuse their towels in the room with no social norm on the sign.
18. Randomized experiment
19. Not all hotels attract the same clientele (rural, urban, resort, business).
20. Researchers wondered if different socially normed messages led to different rates of towel reuse in hotels. Rooms were randomly assigned to have one of five socially normed signs hung on the towel bar. Data were collected on 1,595 instances of potential towel reuse. It was hypothesized that there would be an association between type of message and towel reuse. Data gathered showed 37% towel reuse in the no-normed sign group, 44.1% reuse in the guest-normed group, 49.3% in the room-normed group, 43.4% in the citizen-normed group, and 41% in the gender-normed group. A p-value of 0.024 provided strong evidence against the null and in support of the alternative that there is an association between message type and towel reuse. Follow-up analysis comparing pairs of proportions found that a significantly higher proportion of those in the room-normed group reused their towels than those in the no-social-normed group. Also a significantly higher proportion of those in the room-normed group reused their towels than those in the gender-normed group. I would recommend to the hotel manager that they use the room-normed message to promote towel reuse in their hotel. Over the no-socially-normed

message, this would increase their towel reuse by 4.5%–20.1%. The study should be repeated in a number of different hotels to see if similar results are obtained.

Solutions to Exploration 8.2B

1. Null: The population proportion of nearsightedness is the same in each of the three groups.
   Alt: At least one of the population proportions is different.
2. Null: There is no association between being nearsighted and use of night lights.
   Alt: There is an association between being nearsighted and use of night lights.
3. Yes, there are at least ten observations in each of the six cells of the two-way table.
4. a. $(18 + 78 + 41)/479 = 0.286$
   b. 172
   c. 49.2
   d. 49.2 vs. 18. Quite a bit different, suggesting some evidence against the null hypothesis, because a lot fewer children were nearsighted in this group than we would expect if the null hypothesis were true.
5. Chi-square = 55.52, p-value < 0.0001. We have strong evidence of an association between room darkness and nearsightedness in children.
6. There are a lot less nearsighted children than expected in the dark room group, and a lot more than expected in the room light group.
7. While there are many more nearsighted children in the room light group compared to the dark room group, the proportion of farsighted children in each group is relatively similar.
8. No, the no-association/association way to state hypotheses still works.
9. Yes, there are at least 10 individuals in each cell of the table.
10. It's essentially the same as before. The chi-square statistic is not much different (56.51) and the p-value is still very small (<0.0001).
11. Yes, the overlaid distribution matches the simulated distribution very closely and the p-values are similar. This makes sense because the validity conditions are met.
12. Yes, the additional cell contributions (farsighted cells) are not very large.
13. No, random assignment was not used, so a cause-and-effect conclusion cannot be drawn. The confounding variable here is parental eyesight. Nearsighted parents are more likely to use lights in the room, and, due to genetics, nearsighted parents are more likely to have nearsighted children.